

Guide: A Variable-Length Route Past Some Table Gaps

Collatz proof project — September 5, 2026

Why this matters

The finite table can fail throughout an arbitrarily long initial odd run. That does not mean every merging argument fails there. A classical Garner rule has a length determined by the run itself. We have now formalized its arithmetic applicability and the exact case it leaves open. These identities are known mathematics, also treated explicitly by LaDue (2017).

The two exits

Write an odd seed as $n = 2^a m - 1$, where m is odd. The number a is the length of the initial odd run. After that run the orbit takes an even step. What happens next determines the early comparison with $n - 1$.

If the next step is also even, the two seeds reach the same value after $a + 2$ steps. This works for every run length. If the next step is odd, their values at time $a + 2$ are instead

$$9z + 2 \quad \text{and} \quad z,$$

where z is the value reached from $n - 1$. Lean checks both branches and the precise arithmetic condition selecting them.

Which minimality argument applies

Suppose $n > 1$ were the least positive seed with an unbounded orbit. The smaller seed $n - 1$ would have a bounded orbit. If the two seeds merge, their shared tail makes n bounded too, a contradiction.

This argument needs ordinary merging. It does not require the smaller seed to have noncontracting coefficients. In fact $n - 1$ is even, so its first coefficient is below one. Requiring every replacement to preserve full noncontraction would miss this useful classical rule.

Connection to the previous obstruction

We can construct arbitrarily large members of the table-obstruction family that take the double-even exit. The table only retraces their early odd steps, but the variable-length Garner rule merges them with their predecessor later. We can also construct members taking the single-even exit. The obstruction therefore contains both cases; the new application does not dispose of the whole family.

The remaining gap

In the single-even case, z lies on the bounded orbit of $n - 1$, while $9z + 2$ lies on the supposedly unbounded orbit of n . We have not proved a rule that transfers boundedness from z to $9z + 2$. The exact relationship identifies the unresolved problem; it does not solve it. Collatz remains unproved, including the nontrivial-cycle issue.

How to verify the work

The formal source is `lean/Collatz/OddRunMerges.lean`. From `lean/`, run `lake build Collatz.OddRunMerges`. From the root, run `python3 analysis/odd_run_merges.py` for constructed examples or `python3 -m unittest discover -s analysis -p test_odd_run_merges.py` for independent orbit checks. The generic endpoint identities and least-unbounded exclusion are kernel checked. The CRT application to the obstruction is a written deduction, supported by exact instance checks.